

QUESTION 1

Type: mcq

Prompt:

Which deployment strategy is best for validating a risky release with a small slice of production traffic first?

Guidance:

Pick the safest progressive delivery method.

Rubric:

Strong answers should minimize blast radius before global rollout.

Solution:

Use a canary release because it validates real traffic gradually.

Allow Multiple: false

Options:

- ☐ Big-bang deploy during off-hours
- ☒ Canary release with staged rollout
- ☐ Full blue-green cutover to every user
- ☐ Shadow deploy with no user-facing reads

END QUESTION

QUESTION 2

Type: debug

Prompt:

An API started timing out after a dependency upgrade. Walk through how you would debug it.

Guidance:

Cover telemetry, reproduction, rollback criteria, and how you isolate the change.

Rubric:

Strong answers identify baselines, traces, and mitigation steps.

Solution:

Compare pre/post-upgrade latency, inspect traces, isolate the slow dependency, and define rollback plus instrumentation.

Language: typescript

Starter Code:

```
``ts
export async function fetchAccount(id: string) {
  const profile = await profileClient.get(id);
  const invoices = await billingClient.listInvoices(id);
  return { profile, invoices };
}
``
```

Expected Outcome:

Candidate explains where to instrument, what changed, and how to restore stability safely.

END QUESTION

QUESTION 3

Type: scenario

Prompt:

Design the first 48 hours of an engineering sprint for a reliability fix.

Guidance:

Show milestones, owners, and risk controls.

Rubric:

Look for sequencing, execution realism, and communication.

Solution:

An ideal answer includes triage, implementation, observability, validation, and stakeholder updates.

Deliverable: Execution plan

Constraints:

- One backend engineer and one frontend engineer
- Fix must be observable in production metrics
- Stakeholder update due by end of day two

END QUESTION